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LID AND HYDBRID SUBSTRATE SUPPORT FOR EFFICIENT HEATING AND COOLING IN PROCESS CHAMBERS

Abstract

A lid for a process chamber includes a first ceramic disc including a first dielectric material having a first thermal conductivity, a second ceramic disc including a second dielectric material having a second thermal conductivity, and a bond layer that bonds the second ceramic disc to the first ceramic disc. The first thermal conductivity may be greater than the second thermal conductivity.

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Background/Summary

TECHNICAL FIELD

[0001] Some embodiments of the present disclosure relate, in general, to a hybrid lid and a hybrid substrate support for efficient heating and cooling in process chambers.

BACKGROUND

[0002] Inductively coupled plasma (ICP) is used for etching substrates in semiconductor processes. ICP sources offer several advantages over other plasma sources, including a high plasma density under low working pressures, a controllable ion flux and energy, and a simple configuration. In order to maintain a high plasma density in the ICP, a very high radio frequency (RF) power is often delivered to the coils, which can heat the dielectric lid near the antenna to a high temperature, and cause cracks in the lid. Additionally, when the temperature difference across two points on the lid (also referred to as “temperature delta”) increases beyond a particular difference of (e.g., 15° C.) it may cause thermal stresses in the lid, which may cause the lid to crack and/or break. The probability of such failure increases exponentially beyond a certain temperature delta (e.g., 60° C.).

[0003] Additionally, ICP based chambers use increased source power handling capabilities (~8 to 10 KW). However, existing dielectric lids are composed of alumina, which has low thermal conductivity and high specific heat. Due to low thermal conductivity, the heat transfer efficiency of the lid is low for both cooling and heating purposes. Low thermal conductivity results in high temperature drop across the lid thickness, and a high temperature drop across the thickness results in high thermal stresses, which can cause lids to crack. Additionally, high specific heat of alumina makes it inefficient for both heating and cooling purposes as it requires larger amount of heat to increase or decrease the temperature of lid.

SUMMARY

[0004] Some embodiments of the present disclosure described herein cover a hybrid lid for a process chamber. The hybrid lid includes a first ceramic disc including a first dielectric material having a first thermal conductivity, a second ceramic disc including a second dielectric material having a second thermal conductivity, and a bond layer that bonds the second ceramic disc to the first ceramic disc. The first thermal conductivity may be greater than the second thermal conductivity.

[0005] Some embodiments of the present disclosure described herein cover a lid for a process chamber including a ceramic disc including aluminum nitride. The ceramic disc may further include a first ceramic disc including a first dielectric material having a first thermal conductivity, a second ceramic disc including a second dielectric material having a second thermal conductivity, and a bond layer that bonds the second ceramic disc to the first ceramic disc. The first thermal conductivity may be greater than the second thermal conductivity.

[0006] Some embodiments of the present disclosure described herein cover a hybrid substrate support including a top plate including a first dielectric material having a first thermal conductivity, a bottom plate including a second dielectric material having a second thermal conductivity, and a bond that bonds the top plate and the bottom plate. The first thermal conductivity may be greater than the second thermal conductivity.

[0007] Some embodiments of the present disclosure described herein cover a method including forming a first disc of a lid or a substrate support including a first dielectric material having a first thermal conductivity, forming a second disc of the lid or the substrate support including a second dielectric material having a second thermal conductivity, and bonding the first disc to the second disc. The first thermal conductivity may be greater than the second thermal conductivity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present disclosure is illustrated by way of example, and not by way of limitation, in the

figures of the accompanying drawings in which like references indicate similar elements. It should be noted that different references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references mean at least one.

[0009] FIG. 1 depicts a sectional side view of one embodiment of a processing chamber;

[0010] FIG. 2 depicts a sectional side view of one embodiment of a substrate support in a process chamber;

[0011] FIGS. 3A-3B depict sectional side views of embodiments of a substrate support in a process chamber;

[0012] FIGS. 4A-4D depict sectional side views of embodiments of a hybrid lid of a process chamber;

[0013] FIGS. 5A-5B depict sectional side views of embodiments of a hybrid lid of a process chamber;

[0014] FIGS. 6A-6B depict sectional side views of embodiments of a hybrid lid of a process chamber;

[0015] FIG. 7 illustrates one embodiment of a method of forming a hybrid lid or a hybrid substrate support for a process chamber;

[0016] FIG. 8 depicts a sectional side view of an embodiment of a lid for a process chamber; and

[0017] FIGS. 9A-9B depict sectional side views of an embodiment of a hybrid lid of a process chamber.

DETAILED DESCRIPTION OF EMBODIMENTS

[0018] Embodiments of the present disclosure provide a hybrid substrate support including two or more plates (also referred to as discs). One or more plates of those two or more plates may include one or more functional elements such as chucking electrodes, heating elements, radio frequency (RF) electrodes, zone heaters, pixelated heaters, gas channels, etc. The two or more plates may be bonded using diffusion bonding or metal bonding. Alternatively, a bonding material or an interface layer may be used to bond the two or more plates together. In some embodiments, the hybrid substrate support may include a top plate made of a first material and a bottom plate made of a second material. The first material may have a coefficient of thermal conductivity (simply referred to as thermal conductivity) that is greater than the second material. In some embodiments, the first material may include aluminum nitride, boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The second material may include aluminum oxide (also referred to as alumina) or yttrium oxide (also referred to as yttria), or a heat blocking ceramic.

[0019] The combination of two or more materials used for a hybrid substrate support provides the hybrid substrate support with the ability to withstand higher temperatures. In some embodiments, the top plate has a top portion having a first diameter and a bottom portion having a second diameter, which may be greater than the first diameter. In some embodiments, a bottom surface of the top plate includes a cavity having a third diameter that is less than the first diameter, and a top surface of the bottom plate includes a projection approximately having the third diameter that fits inside the cavity. In some embodiments, an outermost diameter of the bottom plate corresponds to the second diameter. In some embodiments, the bottom plate may include a plurality of holes and the top plate may include a plurality of projections that fit into the holes of the bottom plate.

[0020] Advantages of the hybrid substrate support include a high temperature and low voltage operation because the hybrid substrate support offers a strong chucking force and good thermal uniformity. Good thermal uniformity refers to the even distribution of temperature within a given material. A pure alumina substrate support may develop thermal stresses due to varying temperatures across the thickness or diameter of the substrate support. However, a hybrid substrate support including two or more dielectric materials can lower the thermal gradient, thus improving thermal uniformity across the substrate support. Other advantages include significant energy savings because the alumina in the bottom plate acts as a heat blocking material and so the top plate

can reach the desired temperature with less heat input, thus enabling energy savings (e.g., 20% or more for a given loading condition). Furthermore, the substrate support interface between the electrostatic chuck and the cooling plate experiences lesser temperature (e.g., 50° C. lesser for a given load condition), which helps better manage the interface materials.

[0021] Embodiments of the present disclosure also provide a hybrid lid for a processing chamber. The hybrid lid may include a first ceramic disc including a first dielectric material having a first thermal conductivity, and a second ceramic disc including a second dielectric material having a second thermal conductivity, which may be lower than the first thermal conductivity. The first ceramic disc and the second ceramic disc may be bonded using diffusion bonding or metal bonding. Alternatively, a bonding material or an interface layer may be used to bond the two or more discs together. The bond layer may include at least one of flexible graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy comprising Ni—Ti or Mo—Mg or Cu—Ag or Al alloy. In some embodiments, the first dielectric material may include aluminum nitride, boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The second dielectric material may include aluminum oxide (also referred to as alumina) or yttrium oxide, or a heat blocking ceramic. In some embodiments, the second dielectric material may be spray coated or deposited onto a bottom surface of the first ceramic disc. In some embodiments, the second dielectric material may have a plasma etch resistance that is greater than that of the first dielectric material.

[0022] In some embodiments, the first ceramic disc may include one or more functional elements such as concentric zone heaters. The concentric zone heaters may be individually controlled. Alternatively, the concentric zone heater can be controlled together. In some embodiments, the first ceramic disc forms a top of the lid and has a first diameter. The second ceramic disc forms a bottom of the lid and has a second diameter that is less than the first diameter. In some embodiments, the first ceramic disc forms a top of the lid and has a first diameter at a top of the first ceramic disc and a second diameter at a bottom of the first ceramic disc. The second diameter may be smaller than the first diameter, and the second ceramic disc may have the second diameter. In some embodiments, the second ceramic disc has a first diameter at a top and a second diameter at a bottom of the second ceramic disc that is smaller than the first diameter. The second ceramic disc may include a cavity formed within the top of the second ceramic disc, the cavity having a third diameter that is less than the second diameter. The first ceramic disc may have approximately the third diameter and may be positioned within the cavity. In some embodiments, a bottom surface of the first ceramic disc includes a plurality of concentric grooves, and a top surface of the second ceramic disc includes a plurality of concentric heat transfer fins that fit into the plurality of concentric grooves.

[0023] Embodiments of the present disclosure also provide a method for forming a hybrid lid or a substrate support. The method includes forming a first disc of a lid or a substrate support including a first dielectric material having a first thermal conductivity. The method further includes forming a second disc of the lid or the substrate support including a second dielectric material having a second thermal conductivity. The method further includes bonding the first disc to the second disc. In some embodiments, the first thermal conductivity may be greater than the second thermal conductivity.

[0024] Advantages of the disclosed embodiments include improved thermal stresses on the lid because the hybrid structure results in a lower thermal gradient as compared to a pure alumina lid. A pure alumina lid may develop thermal stresses due to varying temperatures across the lid. However, a hybrid lid including two or more dielectric materials can lower the thermal gradient, thus improving thermal uniformity across the lid. Additionally, the hybrid lid takes 30-40% lesser time to achieve the desired temperature with the same heat loading condition, thus improving thermal efficiency. Because one or more elements of the hybrid lid has a coefficient of thermal conductivity that is greater than alumina, the hybrid lid has better thermal conductivity when

compared to a pure alumina lid. Furthermore, with improved thermal efficiency it takes lesser time for the hybrid lid to cool off during a cooling phase. This results in improved convective cooling of the lid when wafer processing is being carried out.

[0025] FIG. 1 is a sectional view of one embodiment of a processing chamber **100** having a substrate support **150** disposed therein. The processing chamber **100** may be any type of processing chamber, such as a deposition chamber, an etch chamber, an oxidation chamber, an implant chamber, and so on. While the substrate support **150** is described as being an electrostatic chuck assembly or a heater assembly in some embodiments, the substrate support may be replaced with other types of substrate support assemblies, such as a vacuum chuck assembly, a deposition heater assembly, and so on. In one embodiment, the substrate support **150** includes a puck assembly **166** having an upper puck plate bonded to a lower puck plate, as will be discussed in greater detail below. The substrate support **150** may additionally include more than two plates, where each plate may include zero or more different functional elements of the substrate support (e.g., chucking electrodes, radiofrequency (RF) electrodes, main heating electrodes, auxiliary heating electrodes, cooling channels, and so on). The puck assembly **166** can be coupled to a cooling plate by multiple fasteners, or by a bond such as a metal bond, an organic bond, a polymer bond, etc., or a combination thereof.

[0026] The processing chamber **100** includes a chamber body **102** and a lid **104** that enclose an interior volume **106**. Different embodiments of lid **104** will be described in further detail with respect to FIGS. 4A-9B. The chamber body **102** may be fabricated from aluminum, stainless steel, or other suitable material. The chamber body **102** generally includes sidewalls **108** and a bottom **110**. An outer liner **116** may be disposed adjacent the side walls **108** to protect the chamber body **102**. The outer liner **116** may be fabricated and/or coated with a plasma or halogen-containing gas resistant material. In one embodiment, the outer liner **116** is fabricated from aluminum oxide. In another embodiment, the outer liner **116** is fabricated from or coated with yttria, yttrium alloy, or an oxide thereof.

[0027] An exhaust port **126** may be defined in the chamber body **102** and may couple the interior volume **106** to a pump system **128**. The pump system **128** may include one or more pumps and throttle valves utilized to evacuate and regulate the pressure of the interior volume **106** of the processing chamber **100**.

[0028] The lid **104** may be supported on the sidewall **108** of the chamber body **102**. The lid **104** may be opened to allow access to the interior volume **106** of the processing chamber **100**, and may provide a seal for the processing chamber **100** while closed. A gas panel **158** may be coupled to the processing chamber **100** to provide process and/or cleaning gases to the interior volume **106** through a gas distribution assembly **130** or nozzle that may be part of the lid **104**. Examples of processing gases may be used to process in the processing chamber including halogen-containing gas, such as C.sub.2F.sub.6, SF.sub.6, SiCl.sub.4, HBr, NF.sub.3, CF.sub.4, CHF.sub.3, CH.sub.2F.sub.3, Cl.sub.2 and SiF.sub.4, among others, and other gases such as O.sub.2, or N.sub.2O. Examples of carrier gases include N.sub.2, He, Ar, and other gases inert to process gases (e.g., non-reactive gases). The gas distribution assembly **130** may have multiple apertures **132** on the downstream surface of the gas distribution assembly **130** to direct the gas flow to the surface of the substrate **144**. Additionally, or alternatively, the gas distribution assembly **130** can have a center hole where gases are fed through a ceramic gas nozzle. The gas distribution assembly **130** may be fabricated and/or coated by a ceramic material, such as silicon carbide, Yttrium oxide, etc. to provide resistance to halogen-containing chemistries to prevent the gas distribution assembly **130** from corrosion.

[0029] In embodiments, the substrate support **150** is disposed in the interior volume **106** of the processing chamber **100** below the gas distribution assembly **130**. The substrate support **150** holds a substrate **144** during processing. An inner liner **118** may be coated on the periphery of the substrate support **150**. The inner liner **118** may be a halogen-containing gas resist material such as

those discussed with reference to the outer liner **116**. In one embodiment, the inner liner **118** may be fabricated from the same materials as the outer liner **116**.

[0030] In one embodiment, the substrate support **150** is part of a greater assembly **148** that includes the substrate support **150** as well as a mounting plate **162** supporting a pedestal **152**. In one embodiment, the substrate support **150** further includes a thermally conductive base referred to herein as a cooling plate **164** coupled to a puck assembly (also referred to as a puck plate assembly) **166**. The cooling plate **164** may be coupled to the puck assembly **166** by multiple fasteners and/or by a bonding layer. The substrate support **150** described in embodiments may be used for Johnsen-Rahbek and/or Coulombic electrostatic chucking in embodiments. The substrate support **150** may additionally or alternatively be used as a heater, such as a deposition heater that is configured to heat a support substrate **144** during a deposition process.

[0031] In one embodiment, a protective ring **146** is disposed over a portion of the puck assembly **166** at an outer perimeter of the puck assembly **166**. In one embodiment, the puck assembly **166** (or one or more plates of the puck assembly **166**) is coated with a protective layer **136**. Alternatively, the puck assembly **166** may not be coated by a protective layer **136**. The protective layer **136** may be a ceramic such as Y.sub.2O.sub.3 (yttria or yttrium oxide), Y.sub.4Al.sub.2O.sub.9 (YAM), Al.sub.2O.sub.3 (alumina), Y.sub.3Al.sub.5O.sub.12 (YAG), YAlO₃ (YAP), Quartz, SiC (silicon carbide), Si.sub.3N.sub.4 (silicon nitride) Sialon, AlN (aluminum nitride), AlON (aluminum oxynitride), TiO.sub.2 (titania), ZrO.sub.2 (zirconia), TiC (titanium carbide), ZrC (zirconium carbide), TiN (titanium nitride), TiCN (titanium carbon nitride), Y.sub.2O.sub.3 stabilized ZrO.sub.2 (YSZ), and so on. The protective layer may also be a ceramic composite such as Y.sub.3Al.sub.5O.sub.12 distributed in Al.sub.2O.sub.3 matrix, Y.sub.2O.sub.3—ZrO.sub.2 solid solution or a SiC—Si.sub.3N.sub.4 solid solution. The protective layer may also be a ceramic composite that includes a yttrium oxide (also known as yttria and Y.sub.2O.sub.3) containing solid solution. For example, the protective layer may be a ceramic composite that is composed of a compound Y.sub.4Al.sub.2O.sub.9 (YAM) and a solid solution Y.sub.2-xZr.sub.xO.sub.3 (Y.sub.2O.sub.3—ZrO.sub.2 solid solution). Note that pure yttrium oxide as well as yttrium oxide containing solid solutions may be doped with one or more of ZrO.sub.2, Al.sub.2O.sub.3, SiO.sub.2, B.sub.2O.sub.3, Er.sub.2O.sub.3, Nd.sub.2O.sub.3, Nb.sub.2O.sub.5, CeO.sub.2, Sm.sub.2O.sub.3, Yb.sub.2O.sub.3, or other oxides. Also note that pure Aluminum Nitride as well as doped Aluminum Nitride with one or more of ZrO.sub.2, Al.sub.2O.sub.3, SiO.sub.2, B.sub.2O.sub.3, Er.sub.2O.sub.3, Nd.sub.2O.sub.3, Nb.sub.2O.sub.5, CeO.sub.2, Sm.sub.2O.sub.3, Yb.sub.2O.sub.3, or other oxides may be used. Alternatively, the protective layer may be sapphire or MgAlON.

[0032] In embodiments, the puck assembly **166** includes an upper puck plate (not shown) and a lower puck plate (not shown) bonded by a metal and/or organic bond. The puck assembly **166** may also include more than two plates. The upper puck plate and/or one or more other plates may be formed from a monolithic dielectric or electrically insulative material (e.g., having an electrical resistivity of greater than 10^{sup.14} Ohm.Math.meter) that is usable for semiconductor processes at temperatures of 150° C. and above. In one embodiment, the upper puck plate and/or other plate(s) is composed of materials usable from about 20° C. to about 500° C. In one embodiment, the upper puck plate and/or other plate(s) is AlN or another ceramic, or boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The AlN upper puck plate and/or other plate(s) may be undoped or may be doped. For example, the AlN may be doped with Samarium oxide (Sm.sub.2O.sub.3), Cerium oxide (CeO.sub.2), Titanium dioxide (TiO.sub.2), or a transition metal oxide. In one embodiment, the lower puck plate and/or other plate(s) is Al.sub.2O.sub.3. The Al.sub.2O.sub.3 lower puck plate and/or other plate(s) may be undoped or may be doped. For example, the Al.sub.2O.sub.3 may be doped with Titanium dioxide (TiO.sub.2) or a transition metal oxide. In some embodiments, each of the plates of the puck assembly **166** may be formed of a same ceramic. In other embodiments, different plates of the puck

assembly **166** may be formed of different ceramics.

[0033] The lower puck plate and/or one or more other plates may have a coefficient of thermal expansion that is matched or close to a coefficient of thermal expansion of the upper puck plate. In one embodiment, the lower puck plate and/or other plate(s) is a SiC porous body that is infiltrated with an AlSi alloy (referred to as AlSiSiC). The lower puck plate and/or other plate(s) may alternatively be AlN or Al.sub.2O.sub.3 or other ceramic material or a combination thereof (e.g., aluminum oxynitride (ALON)). In one embodiment, the lower puck plate and/or other plate(s) is undoped AlN or undoped Al.sub.2O.sub.3. In one embodiment, the lower puck plate and/or other plate(s) is composed of the same material as the upper puck plate. The AlSiSiC material, AlN or Al.sub.2O.sub.3 may be used, for example, in reactive etch environments or in inert environments.

[0034] In one embodiment, the lower puck plate and/or other plate(s) is Molybdenum.

Molybdenum may be used, for example, if the puck assembly **166** is to be used in an inert environment. Examples of inert environments include environments in which inert gases such as Ar, O₂, N, etc. are flowed. Molybdenum may be used, for example, if the puck assembly **166** is to chuck a substrate for metal deposition. Molybdenum may also be used for the lower puck plate and/or other plate(s) for applications in a corrosive environment (e.g., etch applications). In such an embodiment, exposed surfaces of the lower puck plate and/or other plate(s) may be coated with a plasma resistant coating after the lower puck plate is bonded to the upper puck plate. The plasma coating may be performed via a plasma spray process. The plasma resistant coating may cover, for example, side walls of the lower puck plate and an exposed horizontal step of the lower puck plate. In one embodiment, the plasma resistant coating is Al.sub.2O.sub.3. Alternatively, the plasma resistant coating may be Y.sub.2O.sub.3 or a Y.sub.2O.sub.3 containing oxide. Alternatively, the plasma resistant coating may be any of the materials described with reference to protective layer **136**.

[0035] The mounting plate **162** is coupled to the bottom **110** of the chamber body **102** and includes passages for routing utilities (e.g., fluids, power lines, sensor leads, etc.) to the cooling plate **164** and the puck assembly **166**. The cooling plate **164** and/or puck assembly **166** may include one or more embedded heating elements **176**, embedded thermal isolators **174**, conduits **168**, **170** to control a lateral temperature profile of the substrate support **148**, and/or other functional elements. In embodiments, different functions of the puck assembly **166** may be divided across multiple plates. For example, one plate may include RF electrodes, one plate may include primary heating electrodes, one plate may include auxiliary heating electrodes, and so on. In some embodiments, multiple functions are provided by a single plate. For example, one plate of puck assembly **166** may include both RF electrodes and chucking electrodes. In one embodiment, a thermal gasket **138** and/or o-ring is disposed on at least a portion of the cooling plate **164**.

[0036] The conduits **168**, **170** may be fluidly coupled to a fluid source **172** that circulates a temperature regulating fluid through the conduits **168**, **170**. The embedded thermal isolators **174** may be disposed between the conduits **168**, **170** in one embodiment. The embedded heating elements **176** are regulated by a heater power source **178**. The embedded heating elements **176** may be included in one plate (e.g., top plate) of puck assembly **166**. The conduits **168**, **170** and embedded heating elements **176** may be utilized to control the temperature of the puck assembly **166**, which may heat and/or cool the puck assembly **166** and a substrate (e.g., a wafer) being processed. In one embodiment, the puck assembly **166** includes two separate heating zones that can maintain distinct temperatures. In another embodiment, the puck assembly **166** includes four different heating zones that can maintain distinct temperatures. In other embodiments, the puck assembly **166** includes more than four heating zones (e.g., 8, 16, 32, 64, 128, 216, etc. pixelate heating zones). The temperature of the puck assembly **166** and the thermally conductive base **164** may be monitored using multiple temperature sensors **190**, **192**, which may be monitored using a controller **195**. The temperature sensors **190**, **192** may be included in one plate of puck assembly **166** and/or in multiple plates of the puck assembly **166**, which may be a same plate or plates or

different plate or plates from the plate(s) containing the heating elements **176**.

[0037] The puck assembly **166** may further include multiple gas passages such as grooves, mesas, seal bands, and other surface features that may be formed in an upper surface of a topmost plate of the puck assembly **166**. The gas passages may be fluidly coupled to a source of a heat transfer (or backside) gas, such as He via holes drilled in the plates of the puck assembly **166**. In operation, the backside gas may be provided at controlled pressure into the gas passages to enhance the heat transfer between the puck assembly **166** and the substrate **144**.

[0038] In one embodiment, the puck assembly **166** includes at least one clamping electrode **180** controlled by a chucking power source **182**. The clamping electrode **180** may be included in one plate of puck assembly **166**. The clamping electrode **180** (also referred to as a chucking electrode) may further be coupled to one or more RF power sources **184**, **186** through a matching circuit **188** for maintaining a plasma formed from process and/or other gases within the processing chamber **100**. In one embodiment, a different RF electrode or set of electrodes are connected to one or more RF power sources **184**, **186** and used for maintaining a plasma. The RF electrode(s) may be included in one plate of puck assembly **166**. The one or more RF power sources **184**, **186** may be capable of producing an RF signal having a frequency from about 50 kHz to about 3 GHz and a power of up to about 10,000 Watts. In one embodiment, an RF signal is applied to the metal base, an alternating current (AC) is applied to the heater and a direct current (DC) is applied to the clamping electrode **180**.

[0039] FIG. 2 depicts a cross-sectional view of one embodiment of the substrate support **150**. The substrate support **150** includes the puck assembly **166** and the cooling plate **164** attached to the puck assembly **166**. As shown, an o-ring **214** or gasket may be vulcanized to the cooling plate **164** along a perimeter of a top side of the cooling plate **164**. Alternatively, the o-ring **214** or gasket may be disposed on the top side of the cooling plate **164** without being vulcanized thereto. Alternatively, no o-ring or gasket may be used at the interface of the cooling plate **164** and the puck assembly **166**. Some embodiments are discussed herein with reference to o-rings and gaskets that are vulcanized to at least a portion of the cooling plate **164**. However, it should be understood that the o-rings and/or gaskets may alternatively be vulcanized to the lower puck plate. Alternatively, the o-rings and/or gaskets may not be vulcanized to any surface. In one embodiment, the o-ring **214** or gasket is a perfluoropolymer (PFP) o-ring or polyimide o-ring or gasket. Alternatively, other types of high temperature o-rings may be used. In one embodiment, thermally insulating high temperature o-rings or gaskets are used. The o-ring **214** or gasket may be a stepped o-ring or gasket having a first step at a first thickness and a second step at a second thickness. This may facilitate uniform tightening of fasteners by causing the amount of force used to tighten the fasteners to increase dramatically after a set amount of compression of the o-ring **214** or gasket.

[0040] Additional o-rings (not shown) or gaskets may also be vulcanized to the top side of the cooling plate around a hole **280** at a center of the cooling plate **164** through which cables are run. Other smaller o-rings or gaskets may also be vulcanized to the cooling plate **164** around other openings, around lift pins, and so forth. For example, a gasket (e.g., a PFP gasket or polyimide gasket) may be vulcanized to the top side of the cooling plate **164**. Examples of PFPs usable for the gasket or o-ring **214** are Dupont's™ ECCtreme™, Dupont's KALREZ® and Daikin's® DUPRA™. The o-ring **214** or gasket may provide a vacuum seal between a chamber interior volume and interior volumes within the substrate support **150**. The interior volumes within the substrate support **150** may include open spaces within the pedestal **152** for routing conduits and wiring.

[0041] In one embodiment, the puck assembly **166** has a disc-like shape having an annular periphery that may substantially match the shape and size of the substrate **144** positioned thereon. An upper surface of the puck assembly **166** may have an outer ring **216**, multiple mesas **210**, and channels **212** between the mesas **210**. In one embodiment, the puck assembly **166** includes an upper puck plate **230** bonded to the lower puck plate **232** by a metal bond, a ceramic bond, an organic bond, a polymer bond, or other type of bond. In one embodiment, the bond has different

thermal conductivity in different directions. For example, the bond may have different thermal conductivity in the x, y and/or z directions. In some embodiments, the bond comprises a ceramic with metal fillers (e.g., having ellipsoid particles). The metal fillers may alter a thermal conductivity of the bond in a targeted direction. The thermal conductivity of the bond may accordingly be tailored in one or more directions or planes so that it has isotropic or anisotropic heat transfer properties. In one embodiment, the upper puck plate **230** may be fabricated by an electrically insulative ceramic material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. In one embodiment, the lower puck plate **232** may be fabricated by an electrically insulative ceramic material such as alumina (Al.sub.2O.sub.3), yttrium oxide, or a heat blocking ceramic.

[0042] In one embodiment, the material used for the lower puck plate **232** may be suitably chosen so that a coefficient of thermal expansion (CTE) for the lower puck plate **232** material substantially matches the CTE of the electrically insulative upper puck plate **230** material in order to minimize CTE mismatch and avoid thermo-mechanical stresses which may damage the puck **166** during thermal cycling. In one embodiment, the lower puck plate **232** is Molybdenum. In one embodiment, the lower puck plate **232** is alumina. In one embodiment, the lower puck plate **232** is AlN or Al.sub.2O.sub.3 or yttrium oxide, or a heat blocking ceramic. The lower puck plate may be composed of a different material as the upper puck plate, and may have a different purity level, a different grain size, different amounts of dopants, and so on to provide different material properties for the lower puck plate than the upper puck plate in embodiments.

[0043] The cooling plate **164** attached below the puck **166** may have a disc-like main portion and an annular flange extending outwardly from the main portion and positioned on the pedestal **152**. In one embodiment, the cooling plate **164** may be fabricated by a metal, such as aluminum or stainless steel or other suitable materials. Alternatively, the cooling plate **164** may be fabricated by a composite ceramic, such as an aluminum-silicon alloy infiltrated SiC or Molybdenum to match a thermal expansion coefficient of the puck **166**. The cooling plate **164** should provide good strength and durability as well as heat transfer properties.

[0044] In some embodiments, the upper puck plate **230** and the lower puck plate **232** are bonded together by a bond **250**, which may be a metal bond, an organic bond, a polymer bond, a ceramic bond, or other type of bond. In one embodiment, diffusion bonding is used as a method of metal bonding plates of the substrate support **150** together. In one embodiment, the upper puck plate **230** and the lower puck plate **232** comprise materials which include aluminum (e.g., AlN or Al.sub.2O.sub.3). Bond **250** may be a metal or organic bond **250** that may include an “interlayer” of aluminum foil or other metal foil which is placed in a bonding region between the upper puck plate **230** and the lower puck plate **232**. Pressure and heat may be applied to form a diffusion bond between the aluminum foil and the upper puck plate **230** and between the aluminum foil and lower puck plate **232**. In another embodiment, the diffusion bond may be formed using other interlayer materials which are selected based upon the materials used for upper puck plate **230** and lower puck plate **232**. In another embodiment, the upper puck plate **230** may be directly bonded to the lower puck plate **232** using direct diffusion bonding in which no interlayer is used to form the bond. An organic bond, ceramic bond, polymer bond, or other type of bond may also be formed to bond the plates together.

[0045] In one embodiment, an o-ring **245** is used to protect a metal or organic bond **250**. In one embodiment, no o-ring **245** may be used if a diffusion bond is used to bond the upper puck plate and lower puck plate. A plasma resistant and high temperature o-ring **245** may be made of a perfluoropolymer (PFP) or polyimide in embodiments. The o-ring **245** may be a PFP with inorganic additives such as SiC in an embodiment. The o-ring **245** may be replaceable. When the o-ring **245** degrades it may be removed and a new o-ring may be stretched over the upper puck plate **230** and placed at a perimeter of the puck **166** at an interface between the upper puck plate **230** and the lower puck plate **232**. The o-ring **245** may protect the metal or organic bond **250** from erosion

by plasma. In some embodiments, no o-ring is used to protect the metal or organic bond **250**.

[0046] The upper puck plate **230** includes mesas **210**, channels **212**, and an outer ring **216**. In one embodiment, the upper puck plate **230** includes functional elements such as clamping electrodes **180** and/or one or more heating elements **176**. Alternatively, the clamping electrodes **180** and/or heating elements **176** may be disposed in different plates (e.g., heating elements and/or clamping electrodes may be disposed in lower puck plate **232**). In some embodiments, lower puck plate **232** may include one or more functional elements **220** (e.g., heating elements, clamping electrodes, and/or RF electrodes). The clamping electrodes **180** may be coupled to a chucking power source **182**, and/or to an RF plasma power supply **184** and/or an RF bias power supply **186** via a matching circuit **188**. The upper puck plate **230**, lower puck plate **232** and/or other plates may additionally include gas delivery holes (not shown) through which a gas supply **240** pumps a backside gas such as He. Additionally, the upper puck plate **230**, lower puck plate **232** and/or other plates may additionally include one or more cooling holes (not shown) for a cooling fluid to flow therethrough.

[0047] The upper puck plate **230** and/or lower puck plate **232** may have a thickness of about 1-25 mm or more. In one embodiment, the upper puck plate **230** has a thickness of about 3 mm. The clamping electrodes **180** may be located about 0.25 mm from an upper surface of the upper puck plate **230**, and the heating elements **176** may be located about 1 mm under the clamping electrodes **180**. The heating elements **176** may be screen printed heating elements having a thickness of about 10-200 microns in some embodiments. Alternatively, the heating elements may be resistive coils that use about 1-3 mm of thickness of the upper puck plate **230** in some embodiments. In such an embodiment, the upper puck plate **230** may have a minimum thickness of about 5 mm. In one embodiment, the lower puck plate **232** has a thickness of about 1-25 mm. In some embodiments, the upper and/or lower puck plates have thicknesses ranging from 1 mm to 10 mm, 2 mm to 8 mm, or other thicknesses. In embodiments, different puck plates may have the same or different thicknesses, which may range from 1-25 mm, for example. In embodiments, interface layers or bonding layers between puck plates may have a thickness of about 25 microns to about 1 mm (e.g., 1-14 mil).

[0048] The heating elements **176** may be electrically connected to a heater power source **178** for heating the upper puck plate **230**. The upper puck plate **230** may include electrically insulative materials such as AlN. The lower puck plate **232** and upper puck plate **232** (and/or one or more other plates) may be made of the same materials and/or different materials. In one embodiment, the lower puck plate **232** is made of materials which are different from the materials used for the upper puck plate **230**. The lower puck plate **232** is coupled to and in thermal communication with a cooling plate **164** having one or more conduits **170** (also referred to herein as cooling channels) in fluid communication with fluid source **172**.

[0049] In one embodiment (not shown), the cooling plate **164** includes a base portion. In one embodiment, o-ring **214** may be vulcanized to the base portion. In one embodiment, the cooling plate **164** includes a spring loaded inner heat sink connected to the base portion by one or more springs. The springs apply a force to press the inner heat sink against the puck **166**. A surface of the heat sink may have a predetermined roughness and/or surface features (e.g., mesas) that control heat transfer properties between the puck **166** and the heat sink. Additionally, the material of the heat sink may affect the heat transfer properties. For example, an aluminum heat sink will transfer heat better than a stainless steel heat sink. In one embodiment, the heat sink includes a grafoil layer on an upper surface of the heat sink.

[0050] FIG. 3A depicts a sectional side view of one embodiment of a substrate support **300**.

Substrate support **300** may be used in process chamber **100**, much like substrate support **150**. The substrate support **300** includes a top plate **302** (also referred to as a disc) and a bottom plate **304**. The top plate **302** may be made of a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination

thereof. The bottom plate **304** may include a dielectric material having a thermal conductivity that is less than the thermal conductivity of the top plate **302**.

[0051] In some embodiments, the bottom plate **304** may include alumina (Al.sub.2O.sub.3) or yttrium oxide, or a heat blocking ceramic. The substrate support may further include a bond layer **314** that bonds the top plate **302** and the bottom plate **304**. In some embodiments, the top plate **302** may be diffusion or metal bonded to the bottom plate **304**. In some embodiments, the bond layer **314** may include materials such as graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy including Ni—Ti or Mo—Mg or Cu—Ag or Al alloy.

[0052] The top plate **302** may include one or more functional elements such as a clamp electrode, a heating electrode, a zone heater, a pixelated heater, a radio frequency (RF) electrode, or a gas channel. In some embodiments, the top plate **302** may include one or more chucking electrodes **306** to electrostatically secure the substrate or wafer to the substrate support **300**. In some embodiments, the top plate **302** may include one or more heating elements **308** to heat the top plate **302** to a desired temperature based on process being performed on the wafer or substrate. In some embodiments, the top plate **302** has a top portion having a first diameter **330** and a bottom portion having a second diameter **332**. In some embodiments, the second diameter **332** is greater than the first diameter **330**. In some embodiments, the second diameter **332** is lesser than the first diameter **330**. The bottom surface of the top plate may further include a cavity **336** having a third diameter **334** that may be less than the first diameter **330**. The top surface of the bottom plate **304** may further include a projection **338** approximately having the third diameter **332** that fits inside the cavity **336**. In some embodiments, an outermost diameter of the bottom plate **304** corresponds to the second diameter **332**. The bond layer **314** may be formed in such a way that it conforms to the top surface of the bottom plate **304**. In some embodiments, bond layer **314** may cover an entire area of the top surface of the bottom plate **304**. In some embodiments, bond layer **314** may cover only a portion of the top surface of the bottom plate **304**.

[0053] The substrate support **300** may further include an inner seal band **310** and an outer seal band **312**, which may be alternated with mesas (e.g., mesas **210**) on a top surface of the top plate **302**. Seal bands **310**, **312** can be made from materials that are durable, chemically resistant, and able to withstand extreme conditions of semiconductor processing. In some embodiments, the seal bands **310**, **312** can be made from materials such as elastomers, fluoropolymers, perfluoroelastomers (FFKM), silicone rubber, or Vitron®. The choice of material for the seal bands **310**, **312** depends on the specific requirements of the process in the process chamber, including the chemicals used, the temperature range, the need for electrical insulation, and the vacuum level to be maintained. The material may be selected to ensure they do not outgas or leach contaminants that could compromise the process or the quality of the semiconductor being produced.

[0054] FIG. 3B depicts a sectional side view of one embodiment of a substrate support **350**. The substrate support **350** includes a top plate **322** (also referred to as a disc) and a bottom plate **324**. The top plate **322** may be made of a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The bottom plate **324** may include a dielectric material having a thermal conductivity that is less than the thermal conductivity of the top plate **322**. In some embodiments, the bottom plate **324** may include alumina (Al.sub.2O.sub.3) or yttrium oxide, or a heat blocking ceramic. The substrate support may further include a bond layer **314** that bonds the top plate **322** and the bottom plate **324**. In some embodiments, the top plate **322** may be diffusion or metal bonded to the bottom plate **324**. In some embodiments, the bond layer **324** may include materials such as graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy including Ni—Ti or Mo—Mg or Cu—Ag or Al alloy.

[0055] In the embodiment illustrated in FIG. 3B, the cavity **336** may have chamfered edges instead of sharp edges as illustrated in FIG. 3A. Similarly, the top portion of the bottom plate may have a projection **338** having a profile corresponding to and that fits inside the cavity in top plate **322**. This may allow additional space in the top plate **322** to accommodate additional functional elements.

Additionally, the bottom plate **324** may have a plurality of holes **326**, **328**, and the top plate **322** may include a plurality of projections that fit into the plurality of holes **326**, **328**. A central hole **328** may be used to route one or more terminal wires connecting the one or more functional elements to a power source connection or a ground connection. In some embodiments, the terminal wires may include a DC terminal **318** connecting the chucking electrode to a power source. In some embodiments, the terminal wires may include a heater terminal **320** connecting the heating element(s) to a power source. Through holes **316** may be used to house one or more lift pins that may be used to separate a wafer or substrate from the substrate support **350**. In some embodiments, the lift pins may be made of sapphire, and can be raised to separate a wafer or substrate from the substrate support **350** and lowered to be housed in the holes **316** when not in use.

[0056] FIG. **4A** depicts a sectional side view of an embodiment of a hybrid lid **400** of a process chamber. Hybrid lid **400** may be used in a process chamber **100**, much like lid **104** in FIG. **1**. Hybrid lid **400** may be used for enclosing a process area within the process chamber where one or more operations of a semiconductor manufacturing process may be carried out. Conventional lids are typically made of Al.sub.2O.sub.3 or alumina. In current process chambers, a combination of heating elements, axial fans for forced convection, and a set of baffles for flow direction are used to attain uniform temperature across the chamber lid. The temperature distribution on a lid's surface is dependent on these peripheral heating and cooling elements, and the thermal conductivity and geometry of the material forming the lid. For example, alumina has low thermal conductivity (e.g., about 25 W/mK), and so the temperature distribution of lids formed of alumina may not be uniform. Additionally, with higher source power (e.g., greater than 5 KW) to the process chamber, a temperature delta of greater than 50° C. can be observed on the surface of purely alumina lids. The temperature delta increases the stress at the center of the lid, and directly impacts the “safety factor” of the lid. The safety factor of a lid is directly dependent on the stresses it experiences and if the safety factor falls below a certain range (e.g., due to increased stresses), then the lid may be unsuitable for use in some processes because cracking or breakage may result from using lids below a certain safety factor.

[0057] Accordingly, in some embodiments the hybrid lid **400** includes a top ceramic disc **402** and a bottom ceramic disc **404**. The top ceramic disc **402** may be made of a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The bottom ceramic disc **404** may include a dielectric material having a thermal conductivity that is less than the thermal conductivity of the top ceramic disc **402**. In some embodiments, the bottom ceramic disc **404** may include alumina (Al.sub.2O.sub.3) or yttrium oxide, or a heat blocking ceramic. The top ceramic disc may face away from an interior of a process chamber, and the bottom ceramic disc may face towards the interior of the process chamber. The hybrid lid **400** may further include a bond layer **406** that bonds the top ceramic disc **402** and the bottom ceramic disc **404**. In some embodiments, the top ceramic disc **402** may be diffusion or metal bonded to the bottom ceramic disc **404**. In some embodiments, the bond layer **406** may include materials such as graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy including Ni—Ti or Mo—Mg or Cu—Ag or Al alloy. In some embodiments, the hybrid lid **400** may include a gas distribution assembly **130** or nozzle to allow a fluid or gas to flow therethrough. In some embodiments, the dielectric material of the bottom ceramic disc **404** has a plasma etch resistance that is greater than the plasma etch resistance of the dielectric material of the top ceramic disc **402**. This allows the bottom ceramic disc **404** to be exposed to higher plasma voltages and higher temperatures when the lid is closed. In the embodiment illustrated in FIG. **4A**, the top ceramic disc **402** has a first diameter, and the bottom ceramic disc **404** has a second diameter that is less than the first diameter.

[0058] FIG. **4B** depicts a sectional side view of an embodiment of a hybrid lid **410** of a process chamber. In this embodiment, ceramic disc **412** forms a top of the lid **410** and ceramic disc **414** forms a bottom of the lid **410**. A bond layer **416** may bond the top ceramic disc **412** to the bottom

ceramic disc **414**. The top ceramic disc **412** has a first diameter at a top of the ceramic disc **412** and a second diameter at a bottom of the ceramic disc **412**. In some embodiments, the second diameter is smaller than the first diameter. In some embodiments, the bottom ceramic disc **414** has the second diameter.

[0059] FIG. **4C** depicts a sectional side view of an embodiment of a hybrid lid **420** of a process chamber. In this embodiment, ceramic disc **422** forms a portion of the top of the lid **420** and ceramic disc **424** forms a bottom of the lid **420**. As shown here, the bottom ceramic disc **424** has a first diameter at a top of the ceramic disc and a second diameter at a bottom of the ceramic disc **424** that is smaller than the first diameter. The bottom ceramic disc **424** may include a cavity formed within the top surface. The cavity may have a third diameter that is less than the second diameter. In some embodiments, the top ceramic disc **422** may have approximately the third diameter and may be positioned within the cavity. A bond layer **426** may bond the top ceramic disc **412** to the bottom ceramic disc **414**. In some embodiments, the top ceramic disc **422** and the bottom ceramic disc **424** may be diffusion or metal bonded.

[0060] FIG. **4D** depicts a sectional side view of an embodiment of a hybrid lid **430** of a process chamber. In this embodiment, ceramic disc **432** forms a top of the lid **430**. A bottom disc **434** of the lid **420** can be formed by spraying or depositing a ceramic material onto a bottom surface of the top disc **432**. The top ceramic disc **432** may be made of a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The bottom ceramic disc **434** may include a dielectric material having a thermal conductivity that is less than the thermal conductivity of the top ceramic disc **432**. In some embodiments, the bottom ceramic disc **434** may be formed by spraying or depositing alumina (Al.sub.2O.sub.3) or yttrium oxide on a bottom surface of the top disc **432**.

[0061] FIG. **5A** depicts a top view of an embodiment of a hybrid lid **500** of a process chamber. FIG. **5B** depicts a sectional view of the hybrid lid **500** illustrated in FIG. **5A**. Hybrid lid **500** may be used in a process chamber **100**, much like lid **104** in FIG. **1**. The hybrid lid **500** includes a top ceramic disc **502** and a bottom ceramic disc **504**. The top ceramic disc **502** may be made of a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The bottom ceramic disc **504** may include a dielectric material having a thermal conductivity that is less than the thermal conductivity of the top ceramic disc **502**. In some embodiments, the bottom ceramic disc **504** may include alumina (Al.sub.2O.sub.3) or yttrium oxide, or a heat blocking ceramic. The hybrid lid **500** may further include a bond layer **506** that bonds the top ceramic disc **502** and the bottom ceramic disc **504**. In some embodiments, the top ceramic disc **502** may be diffusion or metal bonded to the bottom ceramic disc **504**. In some embodiments, the bond layer **506** may include materials such as graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy including Ni—Ti or Mo—Mg or Cu—Ag or Al alloy. In some embodiments, the hybrid lid **500** may include a gas distribution assembly **130** or nozzle to allow a fluid or gas to flow therethrough. In some embodiments, the dielectric material of the bottom ceramic disc **504** has a plasma etch resistance that is greater than the plasma etch resistance of the dielectric material of the top ceramic disc **502**. This allows the bottom ceramic disc **504** to be exposed to higher plasma voltages and higher temperatures when the lid is closed.

[0062] In some embodiments, a bottom surface of the top ceramic disc **502** includes a plurality of concentric grooves **510**. A top surface of the bottom ceramic disc **504** includes a plurality of concentric heat transfer fins **508**, which may be used to transfer heat away from the plasma side. The heat transfer fins **508** may be positioned such that areas with stronger plasma have higher heat transfer fins and areas with less strong plasma has lower heat transfer fins. The fins **508** may be made of the same material as the bottom disc **504**. In some embodiments, the fins **508** may be made of a different material from the bottom disc **504**. The grooves **510** may be formed such that each of the concentric heat transfer fins **508** fit into one of the of concentric grooves **510**. The top disc **502** and bottom disc **504** may be diffusion bonded using high temperature and high pressure.

The advantage of diffusion bonding includes the ability to create a high-strength bond without the need for adhesives or mechanical fasteners and generating minimal thermal and mechanical stress on the discs.

[0063] FIG. 6A depicts a sectional side view of an embodiment of a hybrid lid **600** of a process chamber. In this embodiment, ceramic disc **602** forms a top of the lid **600**. A bottom disc **604** of the lid **600** can be formed by spraying or depositing a ceramic material onto a bottom surface of the top disc **602**. The top ceramic disc **602** may be made of a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. The bottom ceramic disc **604** may include a dielectric material having a thermal conductivity that is less than the thermal conductivity of the top ceramic disc **602**. In some embodiments, the bottom ceramic disc **604** may be formed by spraying or depositing alumina (Al.sub.2O.sub.3) or yttrium oxide on a bottom surface of the top disc **602**.

[0064] In some embodiments, the top ceramic disc **602** may include one or more functional elements such as a heating electrode, a zone heater, a pixelated heater, or a gas channel. In some embodiments, the one or more functional elements may include one or more concentric zone heaters, such as inner zone heater **610**, mid-zone heater **608**, and outer zone heater **606**. Each of the heaters may operate at the same temperature or they may operate at different temperatures. For example, inner zone heater **610** may have a lower temperature than mid zone heater **608**, which may have a lower temperature than outer zone heater **606**. Alternatively, inner zone heater **610** may have a lower temperature than mid zone heater **608**, which may have a higher temperature than outer zone heater **606**, or combinations thereof. Each of the heaters **606-610** may be individually controlled by one or more controllers. Alternatively, each of the heaters **606-610** may be collectively controlled by single controller.

[0065] FIG. 6B depicts a sectional side view of an embodiment of a hybrid lid **650** of a process chamber. In this embodiment, ceramic disc **612** forms a top of the lid **650**. A bottom disc **614** of the lid **650** can be formed by spraying or depositing a ceramic material onto a bottom surface of the top disc **612**. The zone heaters **606-610** may be in the form of strips **616-620** instead of wires illustrated in FIG. 6A. Each of the heaters **616-620** may be individually controlled by one or more controllers. Alternatively, each of the heaters **616-620** may be collectively controlled by single controller.

[0066] FIG. 7 illustrates one embodiment of a method **700** of forming a hybrid lid (e.g., hybrid lid **104, 400, 410, 420, 430, 500, 600, 650**) or a hybrid substrate support (e.g., hybrid substrate support **150, 300, 350**) for a process chamber (e.g., process chamber **100**). At block **702**, the method includes forming a first disc of a lid (e.g., disc **402, 412, 422, 432, 502, 602, 612**) or a substrate support (e.g., **230, 302, 322**) from a dielectric material such as aluminum nitride (AlN), boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or a combination thereof. At block **704**, the method includes forming a second disc of the lid (e.g., disc **404, 414, 424, 434, 504, 604, 614**) or the substrate support (e.g., **232, 304, 324**) from a dielectric material having a thermal conductivity that is lesser than that of the first disc. In some embodiments, the second disc may be made of alumina or yttria, or a heat blocking ceramic.

[0067] At block **706**, the first disc and the second disc are bonded together by a bond, which may be a metal bond, an organic bond, a polymer bond, a ceramic bond, or other type of bond. In one embodiment, diffusion bonding is used as a method of metal bonding plates (or discs) of the substrate support together. The bond may be a metal or organic bond that may include an “interlayer” of aluminum foil or other metal foil which is placed in a bonding region between the upper disc and the lower disc. Pressure and heat may be applied to form a diffusion bond between the aluminum foil and the upper disc and between the aluminum foil and lower disc. In another embodiment, the diffusion bond may be formed using other interlayer materials which are selected based upon the materials used for upper disc and lower disc. In another embodiment, the upper disc may be directly bonded to the lower disc using direct diffusion bonding in which no interlayer is

used to form the bond. An organic bond, ceramic bond, polymer bond, or other type of bond may also be formed to bond the plates together.

[0068] It can be challenging to form large AlN sintered ceramic parts due to a large amount of shrinkage that occurs in the sintering process for AlN. For example, sintering an AlN green body into a solid sintered ceramic body of AlN can cause the body to shrink by approximately 20%. This can make it difficult to achieve target tolerances when machining the AlN top plate prior to sintering. Accordingly, in some embodiments the AlN top plate of the hybrid lid is composed of multiple sectional pieces. In some embodiments, the AlN top plate is divided into four equally sized sectional pieces. In some embodiments, the AlN top plate is divided into two, three, four, five, six, seven or eight sectional pieces, which may have the same or different sizes. While a sectionalized top plate is described with reference to AlN, the sectionalized top plate may also be formed of any of the other materials discussed herein with reference to the top plate.

[0069] Each sectional piece of the sectionalized top plate may be smaller than a single monolithic single body top plate, and so may be easier and cheaper to manufacture. Each sectional piece of the AlN top plate may be a pie shaped piece in embodiments. The bottom plate may be a single monolithic plate. Each of the sectional pieces of the AlN top plate may be bonded to the bottom plate using any of the bonding techniques discussed herein. In some embodiments, the sectional pieces are not bonded to one another. Heat generally flows radially from a center of the lid to a perimeter of the lid. Accordingly, failure to bond the sectional pieces to one another may not negatively impact heat transfer properties of the sectional pieces. In some embodiments, the sectional pieces are bonded together.

[0070] FIG. 8 illustrates a cross-sectional view of a portion of a chamber lid **800**, according to some embodiments. The chamber lid **800** may be made entirely from a single dielectric material. In some embodiments, the chamber lid **800** may be made entirely from AlN or aluminum nitride. Chamber lid **800** may have diameter of about 12-30 inches, or more. In some embodiments, the chamber lid **800** may have a diameter of about 26.2 inches.

[0071] Chamber lid **800** may have a thickness of about 1-2 inches, or more. In some embodiments, the chamber lid **800** may have a thickness of about 1.01 inches. Chamber lid **800** may be used to cover the process chamber for a plasma process, and may be exposed to vacuum on the plasma side (e.g., inner surface) and to ambient environment on the outer surface. In some embodiments, one or both surfaces of the chamber lid **800** may be coated (e.g., spray coated) with alumina or another dielectric material. Chamber lid **800** may include one or more heating elements **802**, which may be disposed in the lid or on the outer surface for providing uniform heating across the chamber lid **800**.

[0072] FIG. 9A illustrates an isometric view of a hybrid lid **900**, according to some embodiments. Hybrid lid **900** may be used to cover the process chamber for a plasma process. Hybrid lid **900** may be exposed to vacuum on the plasma side (e.g., inner surface) and ambient environment on the outer surface. Hybrid lid **900** may include two or more ceramic discs **902**, **904** (e.g., a top disc and a bottom disc) that may be bonded using a bonding material. In some embodiments, ceramic discs **902**, **904** may not be bonded together.

[0073] Ceramic disc **904** may be made from a dielectric material such as alumina, which has lower thermal conductivity, and ceramic disc **902** may be made from a dielectric material such as aluminum nitride, which has a thermal conductivity greater than that of alumina. In some embodiments, the thermal conductivity of the aluminum nitride used may be about 50 w/mK or greater. In some embodiments, the thermal conductivity of the aluminum nitride used may be about 75 w/mK or greater. In some embodiments, the thermal conductivity of the aluminum nitride used may be 100 w/mK or greater. In some embodiments, the thermal conductivity of the aluminum nitride used may be 150 w/mK or greater. Hybrid lid **900** may have diameter of about 12-30 inches, or more. In some embodiments, the hybrid lid **900** may have a diameter of about 26.2 inches. Hybrid lid **900** may have a thickness of about 1-2 inches, or more. In some embodiments, the

hybrid lid **900** may have a thickness of about 1.21 inches. In some embodiments, a ratio of the thickness of the ceramic disc **902** to the thickness of ceramic disc **904** may be about 20-100%. In some embodiments, the thickness of the aluminum nitride disc may be about 5 mm, or greater. In some embodiments, the thickness of the alumina disc may be about 1.01 inches, or greater.

[0074] In some embodiments, top ceramic disc **902** may be composed of two or more sections **906**. Although only four sections or quadrants are illustrated in FIG. 3A, ceramic disc **902** may include more than four sections **906**. In some embodiments, top ceramic disc **902** may include two sections **906**. In some embodiments, top ceramic disc **902** may include three sections **906**. In some embodiments, top ceramic disc **902** may include four sections **906**. In some embodiments, top ceramic disc **902** may include eight sections **906**. In some embodiments, top ceramic disc **902** may include ten or more sections **906**. Each of the sections **906** may be attached to the plate **904** using a bonding material, such as graphite, a ceramic such as AlN, an organic elastomer or an epoxy material. In some embodiments, the bonding material may include flexible graphite, an organic elastomer, Al, In, Ni, Ti, AlN, or an alloy comprising Ni—Ti or Mo—Mg or Cu—Ag or Al alloy. Alternatively, or in addition, each of the sections **906** may be attached to the adjacent sections **906** using aluminum nitride. FIG. 9B illustrates a cross-sectional view of a portion of the hybrid lid **900** illustrated in FIG. 9A. Hybrid lid **900** may include a central neck portion **908** where a separate piece of aluminum nitride (or other material having sufficiently high thermal conductivity) may be attached to the bottom lid **904**. The neck portion **908** may be an inner ring that is proximate to a center of the plate **902**. In this neck portion **908**, a ratio of a thickness of the aluminum nitride to a thickness of the alumina in the bottom disc may be 10% or greater. In some embodiments, the thickness of the neck portion **908** may be about 2 mm, or greater. In some embodiments, neck portion **908** may be thinner than sections **906**. For example, sections **906** may have a thickness of about XX and neck portion **908** may have a thickness of about Y.

[0075] The neck portion **908** may be a ring having a U-shaped or L-shaped cross section in embodiments. The neck portion **908** may include a side wall that is taller than a thickness of the sections **906**. In embodiments, the plate **904** may have a ring-shaped recess that is configured to receive the neck portion **908**. When the neck portion **908** is seated in the ring-shaped recess, a side wall of the neck portion **908** may be flush with an upper surface of the sections **906**.

[0076] In embodiments, neck portion **908** is a single ring-shaped piece that is coaxial with plate **904**. A bottom surface of neck portion **908** may be bonded to plate **904**, an outer side wall of neck portion **908** may be bonded to plate **904**, and/or the outer side wall of neck portion **908** may be bonded to each of the sections **906**. The bond between the neck portion **908** and sections **906** may be provided to ensure thermal contact and heat transfer from the neck portion **908** to the sections **906** in embodiments.

[0077] In some embodiments, ceramic discs **902**, **904** may have the same diameter. In some embodiments, ceramic discs **902**, **904** may have different diameters. For example, ceramic disc **902** may have a diameter that is greater than the diameter of ceramic disc **904**. In some embodiments, ceramic disc **902** may have a diameter that is lesser than the diameter of ceramic disc **904**. In some embodiments, the bonding material may include flexible graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy comprising Ni—Ti or Mo—Mg or Cu—Ag or Al alloy, or an epoxy resin.

[0078] Advantages of the hybrid lid include improvement in safety factor of the lid by distributing the temperature more evenly across the surface of the lid. In embodiments where only aluminum nitride is used, aluminum nitride has a higher thermal conductivity when compared to alumina, and so the temperature delta between two points on the surface of the lid is significantly reduced. In embodiments where both aluminum nitride and alumina are used, the bottom disc ensures a tight enclosure for providing an inert environment and the top disc provides even distribution of the temperature across the surface of the lid, thereby resulting in lower temperature deltas and a high safety factor for the overall lid.

[0079] The preceding description sets forth numerous specific details such as examples of specific

systems, components, methods, and so forth, in order to provide a good understanding of several embodiments of the present disclosure. It will be apparent to one skilled in the art, however, that at least some embodiments of the present disclosure may be practiced without these specific details. In other instances, well-known components or methods are not described in detail or are presented in simple block diagram format in order to avoid unnecessarily obscuring the present disclosure. Thus, the specific details set forth are merely exemplary. Particular implementations may vary from these exemplary details and still be contemplated to be within the scope of the present disclosure.

[0080] Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, the appearances of the phrase “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” When the term “about” or “approximately” is used herein, this is intended to mean that the nominal value presented is precise within $\pm 10\%$.

[0081] Although the operations of the methods herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operation may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be in an intermittent and/or alternating manner. In one embodiment, multiple metal bonding operations are performed as a single step.

[0082] It is to be understood that the above description is intended to be illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reading and understanding the above description. The scope of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A lid for a process chamber, the lid comprising: a first ceramic disc comprising a first dielectric material having a first thermal conductivity; a second ceramic disc comprising a second dielectric material having a second thermal conductivity, wherein the first thermal conductivity is greater than the second thermal conductivity; and a bond layer that bonds the second ceramic disc to the first ceramic disc.
2. The lid of claim 1, wherein the first dielectric material comprises at least one of aluminum nitride, boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or combinations thereof, and the second dielectric material comprises aluminum oxide.
3. The lid of claim 1, wherein the first ceramic disc comprises one or more functional elements comprising at least one of a heating electrode, a zone heater, a pixelated heater, or a gas channel.
4. The lid of claim 3, wherein the one or more functional elements comprise one or more concentric zone heaters, wherein the one or more concentric zone heaters are individually controlled.
5. The lid of claim 1, wherein the first dielectric material has a first plasma etch resistance, wherein the second dielectric material has a second plasma etch resistance, and wherein the second plasma etch resistance is greater than the first plasma etch resistance.
6. The lid of claim 1, wherein the bond layer comprises at least one of flexible graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy comprising Ni—Ti or Mo—Mg or Cu—Ag or Al alloy.
7. The lid of claim 1, wherein the first ceramic disc forms a top of the lid and has a first diameter, and wherein the second ceramic disc forms a bottom of the lid and has a second diameter that is less than the first diameter.
8. The lid of claim 1, wherein the first ceramic disc forms a top of the lid and has a first diameter at

a top of the first ceramic disc and a second diameter at a bottom of the first ceramic disc, wherein the second diameter is smaller than the first diameter and wherein the second ceramic disc has the second diameter.

9. The lid of claim 1, wherein: the second ceramic disc has a first diameter at a top of the second ceramic disc and a second diameter at a bottom of the second ceramic disc that is smaller than the first diameter; the second ceramic disc comprises a cavity formed within the top of the second ceramic disc, the cavity having a third diameter that is less than the second diameter; and the first ceramic disc has approximately the third diameter and is positioned within the cavity.

10. The lid of claim 1, wherein: a bottom surface of the first ceramic disc comprises a plurality of concentric grooves; and a top surface of the second ceramic disc comprises a plurality of concentric heat transfer fins, wherein each of the plurality of concentric heat transfer fins fits into one of the plurality of concentric grooves.

11. The lid of claim 1, wherein the first ceramic disc comprises a plurality of pie-shaped sections.

12. The lid of claim 11, wherein the first ceramic disc further comprises a ring shaped section bonded to the plurality of pie-shaped sections at an inner diameter of the plurality of pie-shaped sections.

13. A substrate support, comprising: a top plate comprising a first dielectric material having a first thermal conductivity; a bottom plate comprising a second dielectric material having a second thermal conductivity, wherein the first thermal conductivity is greater than the second thermal conductivity; and a bond that bonds the top plate and the bottom plate.

14. The substrate support of claim 13, wherein the first dielectric material comprises aluminum nitride, boron nitride, silicon carbide, beryllium oxide, sapphire, gallium nitride, zinc oxide, or combinations thereof and the second dielectric material comprises aluminum oxide.

15. The substrate support of claim 13, wherein the top plate comprises one or more functional elements comprising at least one of a clamp electrode, a heating electrode, a zone heater, a pixelated heater, a radio frequency (RF) electrode, or a gas channel.

16. The substrate support of claim 15, wherein the one or more functional elements comprise one or more concentric zone heaters, wherein the one or more concentric zone heaters are individually controlled.

17. The substrate support of claim 15, further comprising: one or more terminal wires connecting the one or more functional elements to at least one of a power source connection or a ground connection.

18. The substrate support of claim 13, wherein the bond comprises at least one of flexible graphite, an organic elastomer, Al, In, Ni, Ti, or an alloy comprising Ni—Ti or Mo—Mg or Cu—Ag or Al alloy.

19. The substrate support of claim 13, wherein the top plate has a top portion having a first diameter and a bottom portion having a second diameter, wherein the second diameter is greater than the first diameter.

20. The substrate support of claim 19, wherein: a bottom surface of the top plate comprises a cavity having a third diameter that is less than the first diameter; a top surface of the bottom plate comprises a projection approximately having the third diameter that fits inside the cavity; and an outermost diameter of the bottom plate corresponds to the second diameter.

21. The substrate support of claim 13, wherein: the bottom plate comprises a plurality of holes; and the top plate comprises a plurality of projections that fit into the plurality of holes.

22. A method comprising: forming a first disc of a lid or a substrate support comprising a first dielectric material having a first thermal conductivity; forming a second disc of the lid or the substrate support comprising a second dielectric material having a second thermal conductivity, wherein the first thermal conductivity is greater than the second thermal conductivity; and bonding the first disc to the second disc.
